

REQUEST FOR PROPOSAL

132/33 kV Main Power Transformer (T1) + 33 kV Earthing/Auxiliary Transformer (T2)

Buyer / EPC contractor: Lighthief Cyprus Ltd (HE 477423)

Project: Grid-connection substation for a utility-scale battery energy storage project, **Cyprus**

Ref: LCY-RFP-TRF-2026 | **Date:** September 2026

Contact: Alexander Papacosta — office@lighthief.com — +357 99 164 158 — solarfarms.cy

Lighthief Cyprus Ltd, as EPC contractor, invites your offer for the two substation transformers described below. Please quote **T1 and T2 as separate line items** and return a signed clause-by-clause compliance statement. A bid covering only T1 is non-compliant.

1. Site & system conditions

Parameter	Value
System	132 / 33 kV; highest system voltage 145 / 36 kV; 50 Hz; load PF 0.9 lag; HV range $\pm 10\%$
Location / delivery port	Cyprus — CIF/DAP Limassol Port
Altitude / ambient	< 300 m; max 50 °C, design 45 °C
Pollution	Heavy / coastal — composite insulators, creepage ≥ 35 mm/kV
Seismic	CYS EN 1998-1, essential facility, importance factor 1.4
Through-fault	132 kV = 31.5 kA / 1 s; 33 kV = 20 kA / 3 s

2. Scope — two transformers (separate line items)

T1 — Main power transformer

- 63 MVA continuous (ONAN/ONAF), 132/33 kV, **YNd11, uk = 21%** (75 °C, CMR, HV base)
- OLTC +12.5% / -18.75%, 25 × 1.25% steps (MR/ABB vacuum type), local/remote/supervisory
- Outdoor, oil-immersed; **Ecodesign Tier 2 (EU 2019/1783)**, PEI compliant; EN 60076 series
- LV insulation: 170 kV BIL / 70 kV AC** (36 kV class); HV: 550 kV BIL / 230 kV AC; HVN 38 kV AC
- 2 cooler banks @ 50% CMR; dual WTI (HV+LV hotspot); online DGA/monitoring; EN 60296 oil; EN 60137 bushings; EN 60214 OLTC; silica-gel breather; flat base + anti-vibration pads; EN 795 anchorage; impact recorders w/ GPS
- State no-load loss, load loss and PEI %.**

T2 — Earthing & auxiliary transformer (separate unit)

- 33 kV / 400–230 V; **zig-zag (interconnected-star) earthing winding** establishing & **solidly earthing** the 33 kV neutral (no NER)
- Earth-fault withstand **≥ 20 kA for 3 s** (or 25 kA for 1 s) per EN 60076-5
- Auxiliary winding ≥ 315 kVA** @ 400/230 V, 3-ph 4-wire (upratable — quote 315 / 400 / 500 kVA, price per step)
- 36 kV class insulation (170/70); EN/IEC 60076 + EN 60289

3. Compliance

Return a **signed clause-by-clause compliance statement** (COMPLY / DEVIATE + note). Explicitly confirm: **YNd11** (not Dyn11), **uk 21%**, **170/70 LV insulation**, **Tier 2 Ecodesign PEI**, and that **T2 is included**. State no-load loss, load loss and PEI % for T1 and T2 — these drive the evaluation.

Open items — how to bid (do not assume)

- **T2 auxiliary kVA:** bid 315 kVA minimum, confirm design is upratable, give price per kVA step.
- **132 kV CTs (T1):** quote 6-CT base **plus a priced option for the 9-CT arrangement** (neutral CT for REF) — final fixed when the connection-bay protection SLD is issued.

4. Delivery (state clearly)

- Ex-works manufacturing lead time (weeks from order/LOI).
- Total delivery time to **Limassol Port, Cyprus** incl. transit + customs clearance.
- **Incoterms 2020:** price **DAP Limassol Port** (baseline); also quote **DDP site (Cyprus)** (inland haulage + offloading).
- Compatibility with an **on-site Q1/Q2 2027** target; transport provisions: impact recorders w/ GPS, lifting/jacking points, shipping oil/nitrogen state, abnormal-load responsibility.

5. Certifications (attach with bid)

- EU Declaration of Conformity + **CE marking**; **EU Ecodesign Tier 2** PEI declaration
- **Independent short-circuit withstand TYPE TEST** to EN 60076-5 (KEMA / CESI / IPH) — calculation-only not accepted
- Temperature-rise (EN 60076-2), impulse/AC withstand (EN 60076-3 at 170/70 LV), sound level (EN 60076-10); routine tests per EN 60076-1; material certs (oil EN 60296, bushings EN 60137, OLTC EN 60214)
- ISO 9001 / 14001 / 45001; EU references for 132 kV / ≥ 63 MVA units; warranty & LTSA terms

6. Commercial & submission

- Prices in **EUR**, itemised: T1, T2, transport/Incoterm options, spares, supervision of erection & commissioning.
- Validity: minimum 90 days.
- **Warranty:** 24 months from commissioning or 36 months from delivery (whichever first); 60 months from delivery preferred. Must cover both T1 and T2.
- Submit to **office@lighthief.com**. Deadline: **Friday 10 October 2026**.
- Format: T1 offer + T2 offer + compliance statement + certificates + commercial offer.

7. Evaluation

1. Technical compliance (T1 **and** T2), incl. genuine Tier 2 Ecodesign & independent short-circuit type test
2. Total cost of ownership (price + capitalised losses/PEI)
3. Delivery time to Limassol vs the Q1–Q2 2027 target
4. Manufacturer track record & EU references; warranty / LTSA